

The Mother of AI Project

Phase 1 — RAG Systems : arXiv Paper Curator

A Learner-Focused Journey into Production RAG Systems

Learn to build modern AI systems from the ground up through hands-on implementation

Master the most in-demand AI engineering skill: **RAG (Retrieval-Augmented Generation)**

Python

3.12+

FastAPI

0.115+

OpenSearch

2.19

Docker

Compose

Status

Week 7 - Advanced Features

What You'll Build

Week 1	Complete infrastructure with Docker, FastAPI, PostgreSQL, OpenSearch, and Airflow
Week 2	Automated data pipeline fetching and parsing academic papers from arXiv
Week 3	Production BM25 keyword search with filtering and relevance scoring
Week 4	Intelligent chunking + hybrid search combining keywords with semantic understanding
Week 5	Complete RAG pipeline with local LLM, streaming responses, and Gradio interface
Week 6	Production monitoring with Langfuse tracing and Redis caching for optimized performance
Week 7	Agentic RAG with LangGraph and Telegram Bot for mobile access

Vue d'ensemble du systeme

Le systeme ingere en continu les nouveaux papiers arXiv (Airflow), les transforme en chunks vectorises stockes dans PostgreSQL et OpenSearch, puis repond aux questions des utilisateurs via une recherche hybride (BM25 + vecteurs), un re-ranking, et un LLM local (Ollama) expose derriere une API FastAPI et une interface Gradio. L'ensemble est observe avec Langfuse (logs, evals RAGAS, feedback, traces) et prepare une future boucle d'entrainement (fine-tuning via Amazon SageMaker).

Le detail complet de l'architecture (flux, etapes numerotees 1 a 12) est represente page 2.

Architecture - Phase 1 : RAG Systems

RESEARCH PAPER CURATOR

